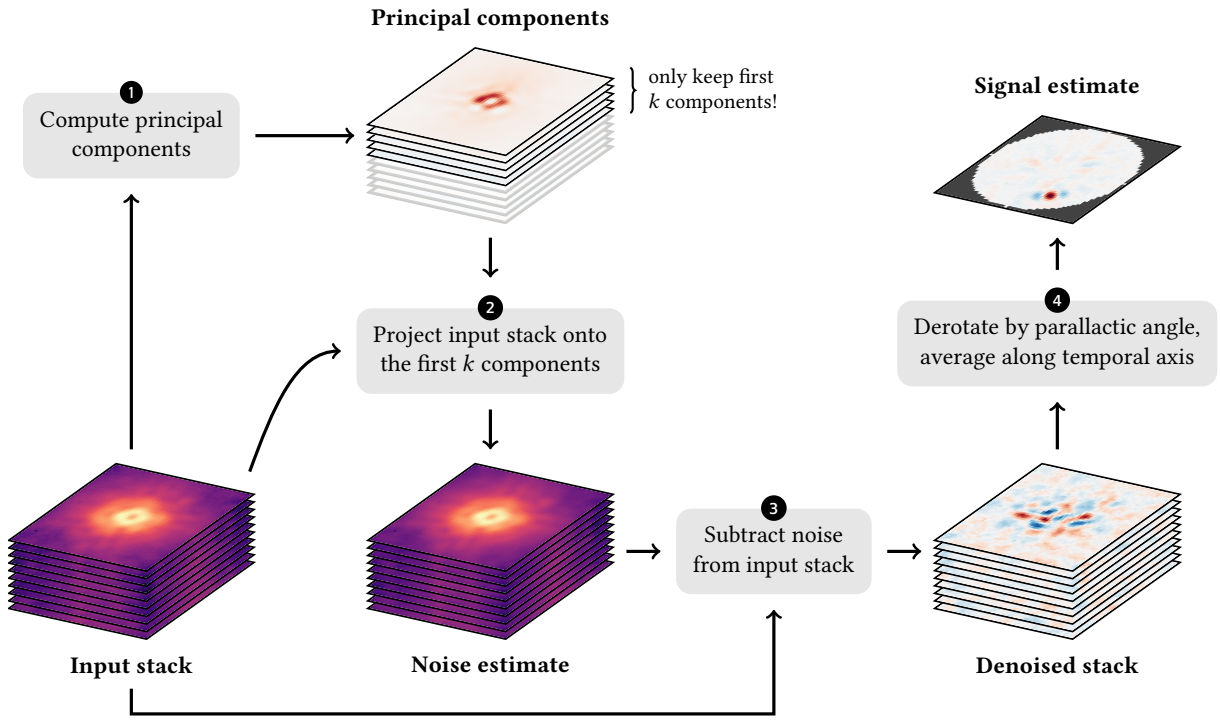

Figure 2.19: Schematic illustration of PCA-based PSF subtraction for ADI. For presentation purposes, different parts of the figure use different color maps. The differences between the input stack and the noise estimate are not visible to the naked eye.

is still widely used both for new discoveries (see, e.g., Mesa et al., 2023, for a recent example) and as a baseline for method development. Second, the algorithm is remarkably simple and easy to explain. Its basic principle is illustrated schematically in fig. 2.19. We outline the application to ADI data here, but transferring the main idea to RDI is straightforward.

The algorithm starts with the "input stack" (i.e., the pre-processed sequence of images) and computes a basis of principal components (or "eigenimages") for it.[63] The principal components (PCs) can be computed efficiently using a singular value decomposition (SVD) assuming that the data are centered, that is, the temporal mean over all frames has been subtracted from each frame.[64] Next, the basis of PCs is truncated to the first k components (in descending order of explained variance). The value of k is the main hyperparameter of the PCA/KLIP algorithm. Its choice depends on the dataset, as well as on the separation of a companion from its host. In the second step, the original input stack is projected onto the PCs, that is, each frame is expressed in the truncated PC basis. This corresponds to the optimal linear dimensionality reduction to k dimensions.[65] The output of this step is our estimate of the PSF. Note that, unlike in median ADI, we get a different estimate for each frame in the input stack. In the next step, the PSF estimate is subtracted from the input stack, and the result is derotated and averaged. Using a

[63] When relying on existing PCA implementations, this may require the images to be flattened into 1D vectors.

[64] Note that this step can already introduce a bias if the mean contains part of the planet signal.

[65] "Optimal" in the sense that this projection minimizes the L_2-distance between the original data and their low-rank approximation; see, e.g., Cunningham & Ghahramani (2015).